

Foundation (Jalapeno)

- Q1.** What is the solution region for the inequality $x + 2y \geq 4$?
- (A) The region below the line
(B) The region above the line
(C) The region to the left of the line
(D) The region to the right of the line
(E) The line itself
- Q2.** A soccer player scores x goals in a game, and his teammate scores y goals. If $x + y \geq 3$, what is the minimum number of goals they must score together?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q3.** The inequality $2x - 3y \leq 5$ can be rewritten as $y \geq$
- (A) $\frac{2}{3}x - \frac{5}{3}$
(B) $\frac{2}{3}x + \frac{5}{3}$
(C) $\frac{3}{2}x - \frac{5}{2}$
(D) $\frac{3}{2}x + \frac{5}{2}$
(E) $\frac{2}{5}x - \frac{3}{5}$
- Q4.** A basketball team has x points from 2-pointers and y points from 3-pointers. If $2x + 3y \geq 20$, what is the minimum number of points they must score?
- (A) 10
(B) 12
(C) 14
(D) 16
(E) 20
- Q5.** The system of inequalities $x + y \geq 2$ and $x - y \leq 1$ has a solution region that is
- (A) Unbounded
(B) Bounded
(C) Empty
(D) Infinite
(E) Undefined
- Q6.** A cyclist travels x miles in a hour, and her competitor travels y miles in an hour. If $x + y \geq 25$, what is the minimum combined speed they must have?
- (A) 20 mph
(B) 22 mph
(C) 25 mph
(D) 28 mph
(E) 30 mph
- Q7.** The inequality $x - 2y \leq 4$ can be graphed on a coordinate plane as
- (A) A solid line with a slope of $\frac{1}{2}$
(B) A dashed line with a slope of $\frac{1}{2}$
(C) A solid line with a slope of $-\frac{1}{2}$
(D) A dashed line with a slope of $-\frac{1}{2}$
(E) A vertical line
- Q8.** A swimmer swims x laps in a pool, and his teammate swims y laps. If $x + y \geq 8$, what is the minimum number of laps they must swim together?
- (A) 6
(B) 7
(C) 8
(D) 9
(E) 10

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \geq 3$ and $x - y \leq 2$ on a coordinate plane. Identify the solution region and provide the coordinates of the vertices.
- Q10.** A football player runs x yards in a game, and his opponent runs y yards. If $x + y \geq 15$ and $x - y \leq 5$, find the minimum and maximum number of yards they can run together. Provide a brief explanation for your answer.

Chef's Tips

- Tip 1: Ensure students understand the concept of graphing inequalities on a coordinate plane.
- Tip 2: Emphasize the importance of identifying the solution region and its boundaries.
- Tip 3: Encourage students to use real-world examples, such as sports statistics, to apply the concepts of systems of linear inequalities.